# Psychometric genetics: a replication of Schwabe (2016)


## Abstract

We partly replicated Schwabe (2016)'s findings that item-level models protect biometric inference from measurement artifacts and that full information improves precision with missing covariates: all 108 reconstructed precision contrasts favor full information, but the broader measurement and power claims are not established. Empirical effects could not be independently refitted, and repeated Bayesian recovery is incomplete. For the illustrative heritability bias, the published positive effect is 0.03315402 versus our 0.03118039; its smaller magnitude does not imply equivalence. All 216 published variance-SE points fall outside conditional comparison bands; unknown source uncertainty and SE implementation prevent a definitive numerical replication verdict.


| Scientific estimand / applicability | Published | Ours | Effect magnitude | Direction | Verdict |
|---|---|---|---|---|---|
| Item-subset invariance; proposition | not established | not established | not established | not established | partial |
| IRT bias remedy; joint claim | not established | not established | not established | not established | not separated |
| Interaction power; design grid | not established | not established | not established | not established | not separated |
| Sum-score artifacts; joint claim | not established | not established | not established | not established | not separated |
| Mathematics AxE slope | 2.0837 | not established | not established | not established | not separated |
| Mathematics AxC slope | 2.6159 | not established | not established | not established | not separated |
| Two response dimensions; proposition | not established | not established | not established | not established | not separated |
| Nine-item subscale; proposition | not established | not established | not established | not established | not separated |
| Liberalism AxE slope | -2.81 | not established | not established | not established | not separated |
| Liberalism AxC slope | 0.54 | not established | not established | not established | not separated |
| AxE sign reversal; joint claim | not established | not established | not established | not established | not separated |
| AxC sign reversal; joint claim | not established | not established | not established | not established | not separated |
| Logvariance identification; proposition | not established | not established | not established | not established | partial |
| Measurement-bias prevention; proposition | not established | not established | not established | not established | not separated |
| Unique finite maximum; proposition | not established | not established | not established | not established | not separated |
| Genetic power adequacy; proposition | not established | not established | not established | not established | not separated |
| Shared-environment power; proposition | not established | not established | not established | not established | not separated |
| 20-item sufficiency; proposition | not established | not established | not established | not established | not separated |
| SES shared-environment slope | -3.0777 | not established | not established | not established | not separated |
| 13 covariate nondetections; joint claim | not established | not established | not established | not established | not separated |
| Missing-data power; grid | not established | not established | not established | not established | not separated |
| Variance-SE ordering; 108 contrasts | not established | not established | not established | not established | partial |
| Observed-data likelihood; proposition | not established | not established | not established | not established | partial |
| MAR A bias; 24 contrasts | not established | not established | not established | not established | partial |
| MAR C bias; 24 contrasts | not established | not established | not established | not established | partial |
| MAR heritability bias; 24 contrasts | not established | not established | not established | not established | partial |
| MCAR A SE: deletion minus full | 0.00106356 | 0.00150235 | not established | same direction | partial |
| MAR heritability: estimate minus truth | 0.03315402 | 0.03118039 | smaller | same direction | partial |

Published values: Schwabe (2016), Tables 3.2, 4.3, 5.4 and Figures 6.3, 6.5. Slopes are log-variance change per source-scale unit; positive means increasing variance. Slope tolerances: 0.00005 (mathematics, SES), 0.005 (liberalism). The last two rows are illustrations, not pooled effects: MCAR, two covariates, 2% missing, A variance SE (squared phenotype units), tolerance 0.00023667095377570213; MAR, five covariates, 10% missing, heritability bias (proportion), tolerance 0.00027219389842495325. Positive means lower full-information SE or upward bias, respectively. Proposition/joint/grid rows have no scalar aggregate or magnitude tolerance. Descriptive point estimates; not equivalence. Source extraction bounds exclude unknown source Monte Carlo uncertainty. Unresolved SE comparability withholds magnitude despite the positive diagnostic contrast. Complete signed components and source locators are linked in Appendix A.

# 1 Introduction

We partly replicated Schwabe's (2016) conclusions about psychometric measurement and missing-covariate analysis, with the strongest numerical evidence supporting the Chapter 6 precision ordering rather than the full dissertation's empirical and power conclusions (Schwabe, 2016). All 108 deletion-minus-full variance-standard-error contrasts were positive with positive multiplicity-adjusted intervals. The illustrative heritability bias was positive in the source (0.03315402) and our reconstruction (0.03118039), and smaller by the geometric tolerance 0.00027219389842495325. Empirical refits, source-repetition Bayesian recovery and direct power comparisons are not established. A blocked comparison is not evidence of nonreplication.

The dissertation asks whether measurement properties can distort genetic and environmental inference, and whether joint modeling of incomplete covariates improves twin analyses. Its 2016 University of Twente version integrates item response theory (IRT) with additive-genetic, shared-environmental and unique-environmental components, denoted A, C and E. It studies latent genetic moderation, observed socioeconomic moderation and incomplete covariates. Chapter 7 qualifies the measurement argument: a common identified scale matters, and item modeling cannot repair poorly informative tails or systematic self-report bias (pp.107–108). We preserve those qualifications.

Our comparison covers seven chapters and Appendices A–G. The 996 linked source records include 26 principal claims and overlapping supporting quantities; they are not 996 independent studies. The first-page table retains all principal claims plus two preselected signed illustrations. Qualitative or compound claims have no invented scalar average. A separate scientific verdict can be partial even when a nonscalar magnitude is not established.

# 2 What we did ourselves, and what we took from others

Our contribution is an independent reconstruction and analytical audit, not a new empirical twin study. Schwabe supplied the model ideas, published designs, numerical summaries, figures and original program specifications. We computed conditional algebraic arguments, generated new fixed-seed data, implemented fits, evaluated convergence, and compared results with source quantities. Printed empirical estimates and digitized figure values remain attributed source evidence.

The Chapter 6 reconstruction comprises 48,000 datasets and 96,000 paired-method fits: 36 primary conditions and 12 separate sensitivity conditions, each with 1,000 datasets. All 96,000 fits passed the recorded numerical guards; none was discarded. Guard checks do not certify a global likelihood optimum. Response-generation checks for Chapters 2 and 5 comprise 2,650 datasets in 19 conditions and no posterior fits. Recomputing standardized reliability on those same datasets adds no independent datasets.

We used R 4.5.0 (R Core Team, 2025), OpenMx 2.21.13 for the Gaussian likelihood (Neale et al., 2016), and lavaan 0.6.19 for independent baseline comparisons (Rosseel, 2012). JAGS 4.3.2 with rjags 4-16 retains the non-Gaussian item-level likelihood (Plummer, 2024); a covariance-only substitute would change it. CODA computes descriptive posterior summaries (Plummer et al., 2006), and posterior 1.6.1 supplies rank-normalized convergence and effective-sample-size diagnostics (Vehtari et al., 2021). Package versions and available execution identities are linked in the artifacts appendix. Ordinary mathematical checks were performed without a Lean certificate.

## 3 Does common measurement imply invariant biometric inference?

A common identified scale supports a restricted population-target result, but it does not guarantee identical finite-data evidence from different item subsets. Under calibrated conditionally independent binary items with positive logistic slopes, two distinct item observations identify the Gaussian latent mean and variance from population marginal and joint probabilities. Cross-twin joint probabilities then identify the covariance through a strictly positive Gaussian covariance derivative (Voigtlaender, 2020). At each supported moderator value,

$$A = 2(K_{\mathrm{MZ}} - K_{\mathrm{DZ}}), \qquad C = 2K_{\mathrm{DZ}} - K_{\mathrm{MZ}}, \qquad E = T - K_{\mathrm{MZ}}.$$

Both zygosities must occur at each of two distinct observed moderator levels to identify the stated log-linear coefficients. These are sufficient mathematical hypotheses, not verified properties of the unavailable applications.

Item subsets can give opposite finite likelihood orderings despite identifying the same population parameter. Consider positive ACE points $(1/4, 1/4, 1/2)$ and $(1/2, 1/8, 3/8)$, both with total variance one. DZ covariance is $3/8$ at both points, whereas MZ covariance increases from $1/2$ to $5/8$. With known Rasch items, one subset containing concordant successes favors the higher MZ covariance; another with success/failure patterns favors the lower covariance. Positive and negative derivatives of their pattern probabilities establish strict opposite orderings. This is not a claim about opposite global fitted optima or significance decisions, nor an assertion that the dissertation intended exact finite-sample equality.

The scope is narrower than universal IRT invariance. Chapter 2's latent-genetic moderation produces a non-Gaussian marginal model not covered by this conditional Gaussian argument. Ordinal and multidimensional observation kernels require their own identification work. Coherent affine changes of latent scale multiply variances by the squared scale factor; heritability is unchanged, but raw variances are not. Nonlinear transformations can change the model family. The complete argument and assumptions are in `handoff/scale-invariance-evidence.json`.

## 4 Empirical interactions and response scales

The original empirical interaction directions cannot be independently confirmed with the available inputs. Mathematics AxE was reported positive at 2.0837 with its 95% highest-posterior-density (HPD) interval excluding zero; AxC was 2.6159 with an interval containing zero (Table 3.2). Liberalism AxE was negative at $-2.81$ and AxC positive at 0.54, both with source HPD exclusions (Table 4.3). The socioeconomic shared-environment slope was $-3.0777$ (Table 5.4). These are source estimates, not our fitted effects. Original NTR/Cito linkage and calibration, Virginia responses, and relevant joint posterior inputs were not recovered by the documented public-source search.

The Chapter 4 dimensionality, nine-item subscale, and IRT-versus-sum-score sign reversals remain blocked empirical targets. The related published conservatism article and supplement provide model and response-category context (Schwabe et al., 2016); they do not supply the missing original responses. Source-only transformations of reported slopes or intervals cannot replace a joint posterior refit. Chapter 6's thirteen covariate nondetections at the reported 99.6% HPD criterion likewise cannot be reconstructed from the simulation grid. Nondetection is not a zero effect or equivalence claim.

Software behavior has a narrower evidential role. BayesTwin's pre-defense revision and version 1.0 were inspected separately (Schwabe, 2017). Its public example is simulated, not dissertation empirical data. Agreement of program semantics or replayed examples does not validate the empirical interaction findings. Each unavailable target, original value and locator is retained in the complete claim inventory.

## 5 Identification, variance functions and likelihood maxima

Log-variance moderation removes particular path-sign ambiguities and is conditionally identifiable, but finite-sample likelihood uniqueness is unresolved. With a shared binary moderator $m \in \{0, 1\}$, the squared path $(a + bm)^2$ has the same endpoint values after $b$ is replaced by $-2a - b$. Even $a > 0$ leaves this reflection unless the endpoint is zero or additional positivity constraints are imposed. For three distinct moderator values, equality of squared affine paths implies equality up to global sign; a strictly positive intercept removes that global ambiguity. The binary and continuous statements therefore need different qualifications (Appendix C).

The exponential variance $V(m) = \exp(\beta_0 + \beta_1 m)$ is positive and has derivative $\beta_1 V(m)$. This proves monotonicity, including the constant zero-slope case, but not concavity of a likelihood. The Chapter 5 hierarchy yields $T = A + C + E$, $K_{\mathrm{MZ}} = A + C$ and $K_{\mathrm{DZ}} = A/2 + C$ under the stated independent innovations; the square-root scaling in Equation 5.14 is essential. Combining these moments with the calibrated observation assumptions establishes population identification and a unique population expected-log-likelihood maximizer.

A finite maximum need not exist for every possible dataset with a free mean. For all-zero binary responses, decreasing the fitted mean improves the likelihood toward an unattained supremum. This defeats a universal existence interpretation, but supplies no example of two finite maxima. The source's statement of "only one maximum" (p.72) is thus not established as an at-most-one finite-maximum result. A proper prior changes the objective; no posterior-unimodality or convergence conclusion follows. The full argument is in `handoff/chapter5-analytical-evidence.json`.

## 6 Missing covariates: precision, bias and power

Full information reproduced the qualitative precision ordering throughout the reconstructed Chapter 6 grid. Three missingness mechanisms, four covariate counts ($q = 2, 3, 4, 5$) and three primary rates (2%, 6%, 10%) supply 36 conditions; all 108 component-specific deletion-minus-full variance-SE contrasts were positive. Paired Student-$t$ intervals with Bonferroni family size 108 excluded zero in every case (Dunn, 1961). For MCAR with $q = 2$ and 2% missingness, the A contrast was 0.00150235, MCSE 0.00002231, and adjusted interval [0.00142398,0.00158072], versus the source-derived 0.00106356.

The literal plotted variance-SE magnitudes were not reproduced within the conditional source-comparison bands. All 216 points lay outside bands combining our Monte Carlo uncertainty with geometric extraction bounds. Original Monte Carlo uncertainty and the source SE implementation are unavailable. Exploratory path-SE comparisons have only 20 points outside their corresponding bands, but path SE is a different estimand and cannot replace variance SE. Accordingly, the headline precision verdict is partial and its magnitude comparison unresolved. The figure's 3% label versus the 2% methods description is retained; 12 additional 3% conditions are a separate sensitivity family.

MAR deletion reproduced the reported bias directions across all 36 component-condition cells. A bias ranged from −0.072831 to −0.006218, C from −0.114958 to −0.007669, and heritability from 0.001416 to 0.031180. Adjusted intervals excluded zero in the expected direction in 10 of 12 A cells, all 12 C cells, and 9 of 12 heritability cells. Full-information biases were much smaller descriptively: A [−0.003192,0.001336], C [−0.002616,0.001017], and heritability [−0.001390,0.002654]. None excluded zero after adjustment, which does not establish unbiasedness or equivalence.

The heritability illustration preserves the contrast and its uncertainty. With MAR, five covariates and 10% missingness, deletion's mean estimate was 0.53118039 against truth 0.5, yielding bias 0.03118039; the source-derived estimate 0.53315402 yielded bias 0.03315402. Our MCSE was 0.00304897 and the ex-

ploratory 72-comparison adjusted interval [0.02080654,0.04155423]. The descriptive magnitude is smaller at the geometric bound 0.00027219389842495325, but this is not evidence of a statistically resolved source difference. Eleven full-information point signs reversed relative to the source; none had an adjusted interval excluding zero. Two conditional MAR source-estimate deviations occur at the ambiguous low-missingness level.

Saved-output checks address 21 additional descriptive Chapter 6 targets without adding fits or new hypothesis tests. All 81 adjacent-covariate-count deletion variance-SE point differences are positive. Absolute C deletion bias exceeds absolute A bias in all twelve MAR cells, with both raw and truth-relative comparisons retained. Under 10% MNAR missingness, full-information E means for two through five covariates are 0.208212, 0.207331, 0.206199 and 0.204763; the last is nearer 0.20 than 0.21. The recommendation-relevant subset has 54 positive pre-existing adjusted precision contrasts, but this does not establish an optimal "more than 3%, more than two covariates" policy. Recovery targets retain biases, MCSEs, RMSEs and between-method differences without inventing an equivalence margin. Five direct-power or unequal-structure targets remain pending.

Precision is not rejection-rate power. The source's general power assertion has no unique quantified alternative, test or rejection rule from which to reconstruct a power comparison. Neither lower SE nor a confidence interval containing zero establishes power or adequacy of twenty items. The Chapter 5 sample-size and item-count claims likewise require appropriate nonzero alternatives and detection fractions, not only posterior standard deviations under null moderation.

The observed-data likelihood and observation kernels were checked separately. The Gaussian covariance construction agrees with Equation 6.5 for $q = 2$–$5$. An exact conditional-variance example gives $2, 3/2, 12/11, 279/286$ as progressively more related observations are supplied, supporting how co-twin information can reduce uncertainty about a missing value. It does not guarantee smaller fitted parameter SE in every dataset. Appendix G's binary kernel uses probabilities 0.001 and 0.999 rather than a deterministic threshold; its continuous observation precision 100 adds variance 0.01 on the standardized scale. These exact semantics are documented in `handoff/chapter6-analytical-evidence.json`.

## 7 Response generation and Bayesian recovery

Generated response properties support some descriptive patterns but do not establish parameter recovery or power. Across 100 datasets each, the floor and ceiling designs had raw reliability means 0.908030 and 0.907533, versus the source descriptor 0.90. Mean skewness was positive 0.534467 and negative $-0.543213$, with the expected sign in all 100 datasets of each design. The source's skewness endpoints concern three estimators on one unknown-seed illustration; our replicate means are not that same estimand. The twelve Chapter 2 power-design generation cells had reliability means rounding to 0.92. The four Chapter 5 60-item designs rounded to 0.90, whereas the 20-item mean 0.738196 rounded to 0.74 against the reported 0.75. Neither favorable rounding nor the number of generated datasets supplies a posterior-fit result.

Changing the reliability formula did not explain the key descriptor gaps. Standardized alpha was 0.906973 for the floor design, 0.906451 for the ceiling design and 0.731098 for twenty items. Their paired standardized-minus-raw differences were $-0.001057$, $-0.001082$ and $-0.007099$ with MCSEs 0.0000570, 0.0000591 and 0.0001094. These compare formulas on the same 2,650 datasets; they are not two independent simulation samples. Source participant subset, aggregation and calibration conventions remain uncertain. All 19 conditions are retained below.

A single-data posterior diagnostic cannot replace the source's 100- or 250-dataset summaries. For one Chapter 2 floor dataset with 500 twin pairs and 60 items, both the original and centered representations completed four chains with 12,000 retained draws per chain. The centered run failed its joint gate:

maximum rank-normalized split/folded $\widehat{R} = 1.01704$, minimum bulk effective sample size 261.0, minimum tail effective sample size 606.2, and maximum mean-MCSE/posterior-SD ratio 0.06217. The failing columns were A, C and E variances and their deterministic logarithms, not six independent failures. The interaction slope's pooled descriptive HPD [−0.64710,0.44688] included zero, but a passing individual diagnostic cannot override inadequate joint mixing. Quantile MCSE is not HPD-endpoint MCSE. These diagnostics follow Vehtari et al. (2021); they do not establish empirical nondetection.

# 8 Further conditional consequences of the models

Three additional analytical results clarify what the source models can support without establishing empirical performance. First, the Gaussian observed-moderator identification argument extends to disjoint MZ and DZ moderator supports if each support is not concentrated on three or fewer values and the same exact log-linear ACE functions apply across zygosities. A nonzero sum of at most four real exponentials has at most three distinct zeros, by an induction using Rolle's theorem. Equal conditional covariance functions at four support points must therefore agree everywhere. The ACE inversion then identifies the coefficients. This is a population result requiring calibrated items and available joint probabilities; it is not a guarantee of stable finite-sample extrapolation.

Second, Chapter 2's non-Gaussian latent model has an explicit fixed-parameter moment inverse. With $v_a > 0$, $v_c \geq 0$, $v_e > 0$ and interaction $\beta$, write

$$\Theta = \mu + A + C + \sqrt{v_e \exp(\beta A)}Z, \quad R = v_e \exp(\beta^2 v_a/2), \quad m_3 = 3v_a\beta R.$$

The twin covariances identify $v_a, v_c$ as above, total variance gives $R = T - v_a - v_c$, and $\beta = m_3/(3v_a R)$, $\log v_e = \log R - \beta^2 v_a/2$. A nonzero interaction gives a nonzero third central moment and therefore a non-Gaussian marginal trait. This inverse identifies latent distributions on a fixed scale; no result shows that finite binary response probabilities identify all these moments. Fixed-parameter moment existence also cannot be exchanged with prior averaging: the source inverse-gamma priors have infinite means.

Third, exact Rasch score likelihood can retain the item model under known calibration. For responses $y_j$, score $S = \sum_j y_j$, and fixed difficulties $b_j$,

$$p(y \mid \theta, b) = \frac{\exp(S\theta - \sum_j b_j y_j)}{\prod_j (1 + \exp(\theta - b_j))}.$$

Summing patterns with the same score replaces $\exp(-\sum b_j y_j)$ by a coefficient depending only on the score and difficulties. Their ratio is independent of every latent-model parameter; integrating the same correlated latent law therefore preserves the normalized posterior. This does not justify a rescaled Gaussian sum-score comparator. Unknown calibration or discarded item masks break the argument, and no computational acceleration or recovery improvement has been measured. These three results are our conditional ordinary derivations, not additional source experiments; complete hypotheses are in `handoff/continuation-evidence-packet.json`.

# 9 Assumptions

## 9.1 Common calibration and observable joint probabilities

The population arguments assume known finite positive logistic slopes, a fixed anchored trait scale, conditional independence and sufficient observable within-person and cross-twin joint probabilities. Unknown calibration, informative item selection or absent joint observations require separate identification arguments. Source-scale invariance is not assumed across arbitrary transformations.

### 9.2 Gaussian shared-moderator reconstruction

The Chapter 5 and Chapter 6 covariance arguments assume the stated Gaussian innovations, common ACE functions across zygosity and known genetic correlations. The continuous-support extension additionally assumes exact log-linear functions across the relevant support. Neither the mixed-data application nor latent-genetic moderation is silently treated as this same Gaussian model.

### 9.3 Underreported generation choices

Chapter 5's reconstructed Bernoulli moderator probability is 0.5, consistent with the stated roughly 11% explained variance but not explicitly specified in the source. Chapter 2's fixed-versus-redrawn difficulty bank and original seeds are unresolved. The generation comparisons therefore test a declared reconstruction. No post-hoc choice of seeds, calibration or equivalence margin is used to force agreement.

### 9.4 Comparison precision and uncertainty

Rounding tolerance is half the source's last printed unit; for derived linear contrasts it is the sum of input bounds. Figure comparisons instead use declared coordinate-extraction bounds. A numerical match would round half up to the source value, close would differ by at most one last-place unit, and mismatch would exceed it, only for a comparable quantity. These labels differ from Monte Carlo uncertainty and reconstruction scope. Student-$t$ Monte Carlo intervals use independent generated datasets with paired method differences; Bonferroni families retain the original dependence and target grouping. The full plan was specified during production, after baseline information was known, rather than wholly preregistered.

## 10 Ambiguities that were resolved

The analytical checks distinguish literal printed formulas from implemented likelihoods; some source inconsistencies remain unresolved rather than being silently repaired.

### 10.1 Chapter2 p21 Eq2.16 worked example

The variance-ratio identity is $\Delta = \exp(\beta_1\sqrt{V_A})$, so $\beta_1 = \log(\Delta)/\sqrt{V_A}$ for positive $V_A$. The neighboring 0.5 versus 0.2 baseline discrepancy does not change this identity; it does prevent treating both worked numbers as one common-baseline example.

### 10.2 Chapter3 p37 Eq3.4

For $p = \text{logit}^{-1}(a(\theta - b))$, information for $\theta$ is $a^2p(1-p)$. At $a = 2, \theta = b$, it is 1, whereas the printed formula gives 0.25. Unit slope resolves the narrow formula, but the source plot implementation is unavailable. Additivity requires conditional independence; inverse square-root information is not an arbitrary posterior SD.

### 10.3 Chapter3 p41 versusp42

The reported 1.63 and 1.85 ratios agree with one genetic standard deviation rather than variance. This identifies the numerical convention without independently recovering the empirical interaction posterior.

### 10.4 Chapter4 p56 Eqs4.6–4.10 versusAppendixB

The literal printed category probabilities sum to 1.25 at zero thresholds, while the Appendix B cumulative-exponential probabilities are normalized and equal 1/3 there. The code uses effective transitions indexed by the second and third thresholds. The normalized implemented target and the erroneous printed equations have different verdicts.

### 10.5 Chapter4 p64

The reported heritability mean 0.54 lies outside the printed HPD interval [0.34,0.51]. The inconsistency is retained; no corrected posterior interval is inferred without the original draws.

### 10.6 Chapter5 p78 versusTable5.1

The table and results use 8,000 pairs while one design sentence says 80,000. The declared reconstruction uses the table design and preserves this discrepancy; the source design cannot be silently declared exact.

### 10.7 Chapter5 p74 versusp84

The separate-moderator program is Appendix E according to the contents and later text; the Appendix F reference is a locator discrepancy. Neither its source code nor this correction supplies extension-power evidence.

### 10.8 Chapter6 pp91–95,101–102;Figs6.3–6.6

The primary reconstruction uses 2% from the methods and a separate 3% sensitivity from plot labels. Both are retained, without selecting the better numerical agreement. The discrepancy affects the low-missingness comparison.

### 10.9 Chapter6 p97 versusTable6.1

The enumerated and tabulated pedagogical categories are six, despite text saying five. The Reformed-liberated/liberal label discrepancy is retained in the empirical source mapping, not treated as a new measured group.

### 10.10 Chapter7 p104 andDutch summaryp117

The simulation includes five covariates as well as two, three and four. Summaries omitting the fifth condition do not justify dropping it from the full grid.

### 10.11 Dutch summaryp116 versusChapter4 abstract/results

The primary Chapter 4 table reports positive shared-environment moderation; the Dutch summary has the opposite wording. The table is the numerical source, and the translation is not a separate experiment.

### 10.12 Chapter6 p97 versusAppendixG pp153–154

Appendix G uses the 0.001/0.999 binary kernel and continuous residual variance 0.01. These are exact probability-model differences from deterministic observation, not evidence that the empirical fit is wrong.

### 10.13 Chapter5 p72 andAppendixC

Binary endpoint sign symmetry, continuous global sign symmetry, population identification and finite-likelihood uniqueness are separate statements. Removing path-sign ambiguity does not prove the last statement. The literal equation checks are supported by handoff/narrow-analytical-evidence.json; the source values and all thirteen ambiguity-to-claim mappings are retained in handoff/continuation-evidence-packet.json.

## 11 What could not be established

The dissertation's broad measurement-remedy, power and empirical conclusions are not settled by the completed narrow checks. Missing original data prevent empirical refits; incomplete repeated posterior experiments prevent source-count recovery and detection fractions; underreported procedures limit exact numerical comparison. Population injectivity does not establish a unique finite-sample optimum. The following limitations retain the complete scientific scope of the evidence.

**Limit 1.** Historical one-dataset posterior runs and short pilots do not reproduce the source100-dataset Chapter2 or250-dataset Chapter5 summaries; convergence and repeated-dataset uncertainty remain unestablished.

**Limit 2.** Unavailable original empirical responses, calibration/linkage records, joint posterior draws and underreported designs prevent the corresponding independent empirical or exact-design comparisons.

**Limit 3.** Figure geometry and arithmetic from printed parameters reuse source evidence; they do not reproduce original data generation or empirical fits.

**Limit 4.** Conditional ordinary mathematical arguments do not supply formal certification or verify empirical calibration, observation and support assumptions.

**Limit 5.** H6.precision-grid:36primary conditions and12additional3%missingness sensitivity conditions are distinct; low-missingness source2%prose/3%plot ambiguity remains. Source-derived geometric bounds are neither original Monte Carlo uncertainty nor equivalence margins.

**Limit 6.** C6-3.MAR-bias.A,C6-3.MAR-bias.C,C6-3.MAR-bias.h2:11point sign reversals occur under full information; none excludes zero after adjustment. Both conditional MAR source-estimate deviations occur at ambiguous low missingness.

**Limit 7.** H2.IRT-bias-remedy,H2.power-grid and Chapter5 Bayesian claims:the bounded four-chain one-dataset diagnostic fails its joint convergence gate. No source100/250dataset aggregate, power fraction or empirical effect is inferred from it.

**Limit 8.** C1.scale-invariance and Chapter5/6 analytical claims:ordinary arguments are assumption-bound and not Lean-certified. Population identification is not finite-sample likelihood uniqueness.

**Limit 9.** Whole-dissertation inventory:996linked records include shared components, source-only arithmetic and parents; they are not996independent findings and no replication success fraction is defined.

**Limit 10.** C2-2.design.alpha, C2-3.alpha, C5-2.alpha.J60 and C5-2.alpha.J20: source alpha formula/aggregation/calibration remain ambiguous; rounding of reconstructed means is not an exact source-estimand verdict. Standardization does not close the floor/ceiling or J20 descriptor gaps.

**Limit 11.** C6-2.SE-trend.MCAR.A and other saved-target assessments: qualitative constancy, slightness, closeness and adequacy lack justified equivalence margins. Descriptive trends cannot settle those qualifications or H6.power-assertion.

**Limit 12.** C5-1.logvariance-identification and C2-1.construction: conditional ordinary proofs do not establish finite-sample maximum existence/uniqueness or observed-response identification of non-Gaussian latent moments.

**Limit 13.** H2.power-grid and H2.IRT-bias-remedy: a failed centered one-dataset joint gate cannot supply source-count recovery or power; final recovery result is withheld pending terminal review.

## 12 Where the reconstruction differs

The reconstruction has limited rather than exact scope wherever seeds, calibration, source aggregation or SE implementation are unknown. Its Gaussian missing-data design is fully fitted, but model guards are not a global optimum certificate. Generation-only comparisons and one-dataset posterior diagnostics do not stand in for the source's repeated recovery experiments. The full copyrighted dissertation and unlicensed original programs are locally inspected sources, not redistributed data. Conditional mathematical extensions add explicit assumptions and do not silently strengthen the dissertation's claims.

## A Principal claims and complete linked inventory

Every source record, condition, original value, attempted calculation, uncertainty statement and remaining target is retained in `handoff/continuation-evidence-packet.json` and `coverage-v4.csv`. The 187 signed component comparisons are stored with their parent effects, including all 108 precision contrasts, 72 MAR bias contrasts and the retained interaction/power components. The 28 first-page rows do not replace them. This inventory is a set of dependent targets, not a replication success denominator. Exact-target accounting is 255 blocked, 391 pending, 29 attempted incomplete, 11 completed, 310 completed under declared reconstruction. In particular, 355 source-count aggregate targets remain pending. A completed diagnostic attempt is not a completed exact source target.

### A.1 Under a common appropriately identified IRT measurement scale, biometric conclusions are asserted to be independent of the particular item subset.

**C1.scale-invariance**: partly replicated. Source: Chapter 1 p5; Chapter 4 p51; Chapter 7 pp107–108. A conditional Gaussian population-target version is supported by an independently reviewed ordinary argument. Literal finite-likelihood invariance does not follow. This does not assert that the dissertation intended exact finite-sample equality, and does not settle its non-Gaussian latent-genetic interaction models.

### A.2 Incorporating an IRT measurement model remedies the scale-induced interaction bias and recovers generating parameters in the studied designs.

**H2.IRT-bias-remedy**: not established. Source: Chapter2 abstract p11; aim p24; discussion p25.. Floor IRT one-dataset beta1=-.1047507 and same-data original-model comparison exist; ceiling IRT source-length recovery and all100-dataset parameter/scenario summaries remain unresolved. Conditional mathematical/software checks cannot establish the full conjunction.

### A.3 The proposed method has useful power to detect genotype–environment interaction under the investigated sample-size and effect-size conditions.

**H2.power-grid**: not established. Source: Chapter2 abstract p11; results p22; discussion p25.. All12 design cells have short one-chain pilots, not100-dataset detection fractions; curvilinear extension design unavailable. No component success fraction or single pooled power is meaningful.

### A.4 Sum-score analysis can produce spurious genotype–environment interaction from measurement-scale properties.

**H2.sumscore-artifacts**: not established. Source: Chapter2 abstract p11; results pp21–22; discussion pp24–25.. Historical source-length one-dataset floor/ceiling/symmetric sumscore beta1 means1.0951643,-.8566462,.1003354 supply diagnostic patterns only. Source100-dataset full constituent summaries and convergence are incomplete; no pooled artifact magnitude or component match fraction used.

### A.5 Unique-environmental variance increases with genetic mathematics ability; the AxE95%HPD excludes zero.

**C3-4.3.2.beta1**: blocked. Source: Chapter 3 p42, Table3.2. Independent target C3-4.3.2.beta1 requires original NTR/Cito/SES responses, linkage/calibration or posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.6 The AxC point estimate is positive but its95%HPD includes zero; shared-environment moderation is not detected.

**C3-4.3.2.gamma1**: blocked. Source: Chapter 3 p42, Table3.2. Independent target C3-4.3.2.gamma1 requires original NTR/Cito/SES responses, linkage/calibration or posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.7 The28-item scale measures two distinct dimensions: liberalism–conservatism and use of the question-mark response.

**C4-2.two-dimensions**: blocked. Source: Chapter 4 abstract p47, pp58–61,Figs4.1–4.4. Independent target C4-2.two-dimensions requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.8 Nine selected items constitute the proposed ordinal social-liberalism subscale.

**C4-3.subscale**: blocked. Source: Chapter 4 pp62,66. Independent target C4-3.subscale requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.9 Unique-environmental variance decreases with genetic liberalism; the negative AxE95%HPD excludes zero.

**C4-5.4.3.beta1**: blocked. Source: Chapter 4 p63,Table4.3. Independent target C4-5.4.3.beta1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.10 Shared-environmental variance increases with genetic liberalism; the positive AxC95%HPD excludes zero.

**C4-5.4.3.gamma1**: blocked. Source: Chapter 4 p63,Table4.3. Independent target C4-5.4.3.gamma1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.11 The interaction point-estimate direction changes between IRT and sum-score analyses: beta1

**C4-5.reversal.beta1**: blocked. Source: Chapter 4 pp63–64,66–67,Tables4.3–4.4. Independent target C4-5.reversal.beta1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.12 The interaction point-estimate direction changes between IRT and sum-score analyses: gamma1

**C4-5.reversal.gamma1**: blocked. Source: Chapter 4 pp63–64,66–67,Tables4.3–4.4. Independent target C4-5.reversal.gamma1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.13 The alternative logvariance ACE×M model is asserted to be uniquely identified.

**C5-1.logvariance-identification**: partly replicated. Source: Chapter 5 pp69,72–77,83–84; AppendixC pp138–139. Conditional theorem only; empirical calibration/observation/support assumptions unverified and faithful Lean bridge absent.

### A.14 Simultaneous IRT and ACE×M modelling is asserted to prevent spurious moderation arising from heterogeneous measurement error.

**C5-1.measurement-bias**: not established. Source: Chapter 5 pp69,72–77,83–84; AppendixC pp138–139. A general prevention claim is not established by single-dataset Chapter2 diagnostics or conditional identification; further model/design checks remain feasible.

### A.15 The source states the logvariance model has only one maximum in its likelihood function.

**C5-1.one-likelihood-maximum**: not established. Source: Chapter 5 pp69,72–77,83–84; AppendixC pp138–139. Partially addressed; finite-sample at-most-one-maximum claim remains unresolved. Do not label it disproved solely by a no-finite-maximum example, and do not infer it from population injectivity.

### A.16 Large samples are said to be necessary to detect beta1a moderation.

**C5-2.power-inference.beta1a**: not established. Source: Chapter 5 abstract p69,pp78,83. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or20-item sufficiency. Feasible assessment remains pending.

### A.17 Large samples are said to be necessary to detect beta1c moderation.

**C5-2.power-inference.beta1c**: not established. Source: Chapter 5 abstract p69,pp78,83. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or20-item sufficiency. Feasible assessment remains pending.

### A.18 As few as20phenotypic items are asserted sufficient to find moderation effects.

**C5-3.twenty-sufficient**: not established. Source: Chapter 5 pp80,83. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or20-item sufficiency. Feasible assessment remains pending.

### A.19 Shared-environmental variance is lower for high-SES families; the negative C×SES95%HPD excludes zero.

**C5-5.5.4.beta1c**: blocked. Source: Chapter 5 p82,Table5.4. Independent target C5-5.5.4.beta1c requires original NTR/Cito/SES responses, linkage/calibration or posterior inputs absent after documented

public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

### A.20 None of the thirteen measured covariate coefficients was detected in the educational-achievement application using the reported99.6%HPD criterion.

**H6.covariate-nondetection**: blocked. Source: Chapter6 abstract p86; results p99 and Table6.3 p100; discussion p102.. The original linked NTR/Cito educational-achievement data and original fitted-model inputs were unavailable after documented permitted acquisition. Independent empirical counts, percentages, fitted estimates and nondetection results cannot be recovered from the Chapter6 simulations.

### A.21 The source asserts that full information is more powerful than deletion regardless of the investigated missingness mechanism, covariate count or amount of missingness.

**H6.power-assertion**: not established. Source: Chapter6 abstract pp85–86; pp91,101–102; Chapter7 p104.. Lower estimated SE is not rejection-rate power. Chapter6 provides no unique nonzero alternative, test/rejection rule or source power fractions. This production grid estimates precision and bias, not a direct power experiment; no new test definition is invented.

### A.22 Full information gives lower variance-component standard errors than deletion across the investigated missingness mechanisms and design grid.

**H6.precision-grid**: partly replicated. Source: Chapter6 results pp93–95; discussion p101.. All 108 signed deletion-minus-full variance-SE differences are positive and all 108 Bonferroni-adjusted intervals exclude zero positively. The precision ordering is reproduced in the complete primary grid. Published literal variance-SE points differ from our estimates in all 216 conditional comparison bands; scientific comparability remains unresolved because original SE implementation and original Monte Carlo uncertainty are unavailable. Path SE is an exploratory different estimand. This does not establish rejection-rate power.

### A.23 The full-information model uses all observed phenotype and covariate components when covariate data are incomplete.

**C6-1.observed-likelihood**: partly replicated. Source: Chapter 6 pp88–91;AppendixF. The specified Gaussian construction and computational realization are supported. This restricted model-level result does not establish correct population distribution, empirical data validity or universal inference improvements.

### A.24 MAR deletion underestimates A variance while full information remains close to the generating value.

**C6-3.MAR-bias.A**: partly replicated. Source: Chapter6 pp94–95,Fig6.5. Deletion bias is negative in12/12conditions, with10/12 adjusted intervals excluding0 in that direction; bias range [-0.0728311524251116,-0.0062175646058198804]. Full-information bias range [-0.0031917638748272703,0.001335885122
with0/12 adjusted intervals excluding0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing0.

## A.25 MAR deletion underestimates C variance while full information remains close to the generating value.

**C6-3.MAR-bias.C**: partly replicated. Source: Chapter6 pp94–95,Fig6.5. Deletion bias is negative in12/12conditions, with12/12 adjusted intervals excluding0 in that direction; bias range [-0.11495778041479585,-0.007668582657040717]. Full-information bias range [-0.002616392569283762,0.00101714527283
with0/12 adjusted intervals excluding0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing0.

## A.26 MAR deletion overestimates heritability while full information remains close to the generating value.

**C6-3.MAR-bias.h2**: partly replicated. Source: Chapter6 pp94–95,Fig6.5. Deletion bias is positive in12/12conditions, with9/12 adjusted intervals excluding0 in that direction; bias range [0.001416215315186544,0.03118
Full-information bias range [-0.001390288356138391,0.0026536390348624117], with0/12 adjusted intervals excluding0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing0.

## A.27 Coverage by source section

The section totals below may overlap because an assertion can be used in multiple source locations. Repeated conclusions, cited background and future proposals are not counted as new experiments.

| Source section | Coverage and exclusions |
|---|---|
| chapter_1; PDF pages10–18; printed pages1–9 | 7 linked claims. Cited textbook and external empirical results are background. Repeated dissertation conclusions retain the originating chapter claim IDs rather than being counted as new experiments. |
| chapter_2; PDF pages20–36; printed pages11–27 | 223 linked claims. All source-linked targets retained. |
| chapter_3; PDF pages38–55; printed pages29–46 | 54 linked claims. All source-linked targets retained. |
| chapter_4; PDF pages56–77; printed pages47–68 | 98 linked claims. All source-linked targets retained. |
| chapter_5; PDF pages78–93; printed pages69–84 | 206 linked claims. All source-linked targets retained. |
| chapter_6; PDF pages94–111; printed pages85–102 | 408 linked claims. All source-linked targets retained. |

| Source section | Coverage and exclusions |
|---|---|
| chapter_7; PDF pages112–123; printed pages103–114 | 8 linked claims. Cited textbook and external empirical results are background. Repeated dissertation conclusions retain the originating chapter claim IDs rather than being counted as new experiments. |
| appendix_A; PDF pages140–142; printed pages131–133 | 3 linked claims. All source-linked targets retained. |
| appendix_B; PDF pages143–146; printed pages134–137 | 8 linked claims. All source-linked targets retained. |
| appendix_C; PDF pages147–148; printed pages138–139 | 4 linked claims. All source-linked targets retained. |
| appendix_D; PDF pages149–151; printed pages140–142 | 1 linked claims. All source-linked targets retained. |
| appendix_E; PDF pages152–155; printed pages143–146 | 2 linked claims. All source-linked targets retained. |
| appendix_F; PDF pages156–158; printed pages147–149 | 2 linked claims. All source-linked targets retained. |
| appendix_G; PDF pages159–163; printed pages150–154 | 3 linked claims. All source-linked targets retained. |
| frontmatter; PDF pages1–9 | 0 linked claims. Degree/title/edition metadata and table of contents; no original scientific result. |
| dutch_summary; PDF pages124–126 | 0 linked claims. Translated summaries duplicate mapped Chapter2–6 claims; no new experiment. Wording discrepancies retained in ambiguities. |
| bibliography; PDF pages128–136 | 0 linked claims. Cited prior works are background; their scientific results are not original dissertation replication targets. |
| acknowledgements; PDF pages138–138 | 0 linked claims. Acknowledgements of collaborators/funding, not original scientific results. |
| backcover; PDF pages164–164 | 0 linked claims. Print production marks, not a scientific result. |

# B Generation comparisons in all 19 conditions

Published alpha descriptors come from Chapters 2 and 5 of Schwabe (2016); raw and standardized estimates below are our dataset means. Unknown source aggregation withholds exact-source numerical verdicts. The paired formula comparison changes no underlying observations. Condition numbering links to the complete scientific packet.

| Cell | Chapter | Pairs | Items | Source alpha | Raw alpha | Standardized |
|---|---|---|---|---|---|---|
| 1 | 2 | 500 | 60 | 0.90 | 0.908030 | 0.906973 |
| 2 | 2 | 500 | 60 | 0.90 | 0.907533 | 0.906451 |
| 3 | 2 | 500 | 60 | 0.92 | 0.915498 | 0.914910 |
| 4 | 2 | 1000 | 60 | 0.92 | 0.915660 | 0.915071 |
| 5 | 2 | 2000 | 60 | 0.92 | 0.915653 | 0.915039 |
| 6 | 2 | 500 | 60 | 0.92 | 0.915750 | 0.915064 |
| 7 | 2 | 1000 | 60 | 0.92 | 0.916238 | 0.915626 |
| 8 | 2 | 2000 | 60 | 0.92 | 0.915894 | 0.915258 |
| 9 | 2 | 500 | 60 | 0.92 | 0.916459 | 0.915872 |
| 10 | 2 | 1000 | 60 | 0.92 | 0.916481 | 0.915926 |
| 11 | 2 | 2000 | 60 | 0.92 | 0.916197 | 0.915574 |
| 12 | 2 | 500 | 60 | 0.92 | 0.917088 | 0.916535 |
| 13 | 2 | 1000 | 60 | 0.92 | 0.916215 | 0.915631 |
| 14 | 2 | 2000 | 60 | 0.92 | 0.917012 | 0.916441 |
| 15 | 5 | 1000 | 60 | 0.90 | 0.897739 | 0.894292 |
| 16 | 5 | 2000 | 60 | 0.90 | 0.897882 | 0.894467 |
| 17 | 5 | 4000 | 60 | 0.90 | 0.897706 | 0.894287 |
| 18 | 5 | 8000 | 60 | 0.90 | 0.897792 | 0.894389 |
| 19 | 5 | 1000 | 20 | 0.75 | 0.738196 | 0.731098 |

## C Availability ledger

The dated public-source search below describes what was obtained, not proof that unavailable inputs do not exist. The ledger is copied from `handoff/scientific-availability.json`.

| Date and source | Result |
|---|---|
| 2026-10-30 Institutional final-version record | Verified PhD, institution, advisers, date, DOI, ISBN and final-version designation. |
| 2026-10-30 Author dissertation GitHub repository | Full164-page thesis and five model programs obtained at6c2c48901209fc76ba932bb094c61abba5ef160d. No raw empirical datasets or Monte Carlo driver/seeds. Full text read. Original files kept local; rights reserved/no code license. |
| 2026-10-30 All12 public author GitHub repositories | All cloned; immutable hashes in author-repositories.json. Data-file and code audit found workshop data, BayesTwin simulated posterior example and unrelated projects; no matching NTR/Cito/Virginia study inputs or curvilinear simulation driver. This is a search limit, not proof of absence. |

| Date and source | Result |
|---|---|
| 2026-10-30 OSF indexed search | Queries site:osf.io Schwabe twin and exact Inga Schwabe2016 found later unrelated projects/preprints, no matching dissertation simulation or empirical input. No identified thesis OSF project. |
| 2026-10-30 GitLab indexed search | Queries Schwabe and exact ingaschwabe returned unrelated authors/no matching repository. No claim of exhaustive private-site search. |
| 2026-10-30 Zenodo indexed search | Queries Schwabe twin and exact Inga Schwabe returned unrelated records/later projects; no matching dissertation data/code record. |
| 2026-10-30 Chapter4 article and supplement | Publisher confirms CC BY4.0 article and links ten-page supplement. Supplement accessible through browser; contains category plots, fit plots and JAGS program, not raw responses. Original thesis code replayed locally. Supplement findings cannot be refitted without Virginia30K responses. |
| 2026-10-30 BayesTwin repository history | Current code656afd72d45ecd13f54b72c9400cceface8d423f, version1.0 dated2017, GPL¿=2. Fetched full history; last pre-defense revision a8e5d6b5ee6a8c76bdd0234a00fbcc809712ac08 dated2015-10-15 available for separate replay. Example results.rda explicitly simulated, not dissertation empirical data. |
| 2026-10-30 Historical statistical engine specification | Thesis cites R/rjags2013 and OpenMx2011/JAGS2003 publications but no complete sessionInfo, engine patch versions, random seeds or driver. Exact historical environment not determined. Current pinned versions and original-model replay used with explicit scope limit. |

## D Scientific compliance ledger

The following ledger is copied from `handoff/continuation-scientific-compliance.json`. Departures qualify the scope of the scientific evidence.

| Requirement | Status | Evidence and justification |
|---|---|---|
| Whole-dissertation coverage | followed | coverage-v4.csv and handoff/continuation-evidence-packet.json cover 19 sections, 996 dependent source records, 26 headline claims and 187 effect components. Exact-target work: 255 blocked, 391 pending, 29 attempted-incomplete, 11 completed and 310 completed under declared reconstruction. Coverage is not whole-study replication. |
| Original-source and input audit | followed | handoff/scientific-availability.json records dated permitted acquisition attempts. All12public author repositories were examined. Unavailable empirical inputs and simulation specifications retain unresolved targets. |

| Requirement | Status | Evidence and justification |
|---|---|---|
| Preferred statistical engines | followed | provenance/preflight.json records R4.5.0,lavaan0.6.19 andOpenMx2.21.13. OpenMx implements the Chapter6 Gaussian likelihood; outputs/engine-agreement.csv supplies independent baseline checks. |
| JAGS for non-Gaussian latent item models | departed | A covariance-only SEM would change the source likelihood. JAGS4.3.2/rjags4-16 reconstructs the item-level models; posterior1.6.1 computes diagnostics. Independent generator, baseline engine and model-semantic checks do not independently validate the full posterior. The continuation packet embeds the original and centered four-chain diagnostic evidence and their scientific limits. JAGS preserves the non-Gaussian item-level likelihood. |
| Original repetition counts and adequate posterior convergence | departed | Chapter 6 completes 1000 datasets in 36 primary and 12 sensitivity conditions, 96000 method fits. Response generation completes 2650 datasets across 19 Chapter 2/5 conditions; alpha sensitivity reuses those datasets with zero additional independent datasets and zero posterior fits. Source-count Bayesian recovery/power aggregates remain incomplete. Both completed original and centered 12000-retained-draw-per-chain diagnostics fail their joint convergence gate. Final recovery outcome awaits terminal independent review. handoff/continuation-evidence-packet.json: continuation.scientific_evidence and current_coverage; final recovery reservation. |
| Signed estimands, uncertainty and multiplicity | followed | production/analysis-plan.json separates variance/pathSE, deletion-minus-full contrasts, estimate-minus-truth bias and source deviations. Geometric extraction margins, ourMonteCarlo error and unknownsourceMCerror remain distinct; descriptive magnitude is not equivalence. |
| Analytical verification and formal proof | departed | Reviewed ordinary mathematical arguments and exact algebra checks are available; no Lean certificate is claimed. Conditional population identification, fixed-parameter latent moment inversion and exact Rasch score-likelihood sufficiency do not establish finite-likelihood uniqueness, non-Gaussian observed-item identification, empirical recovery, power or convergence. Absence of formal certification does not prevent reporting a scoped ordinary proof. Continuation packet embeds the ordinary proofs, assumptions, source locators and reviewed scientific limits. |
| Source attribution and permitted redistribution | followed | Verified replicated_work identity, source_pdf/source_text hashes and claim-level source locators in the continuation packet. Dissertation and original programs remain local; source attribution must be preserved in new writing. |

| Requirement | Status | Evidence and justification |
|---|---|---|
| Reproducible versions and commands | followed | handoff/terminal-reproduction.md specifies immutable completed-analysis commit and commands; README.md documents runtime, simulation and diagnostic execution. Navigation links to master are separate from immutable reproduction revisions. |

# E Artifacts and reproduction

The links below identify the actual filenames on the repository's master branch; they are navigation links rather than immutable versions. The completed Chapter 6 saved-report reproduction is pinned separately to commit `a9e7bd4a70cfa3a40f89d3525efd0dd98a5f033a`. From `formalization/schwabe-psychometric-genetic` run the commands in `handoff/terminal-reproduction.md`. They require Python, NumPy and SciPy and write fresh output directories; they rebuild analysis from saved reports without starting fits. The packet's hashes identify the exact evidence used here.

| Artifact | Scientific role |
|---|---|
| handoff/continuation-evidence-packet.json | All 996 claims, all signed effects, limits and source hashes |
| coverage-v4.csv | Exact-target and attempted-work status by source record |
| production/chapter6-v1/terminal-analysis/signed-SE-contrasts.csv | Paired variance and exploratory path SE contrasts |
| production/chapter6-v1/terminal-analysis/signed-bias-contrasts.csv | Signed MAR biases and uncertainty |
| production/chapter6-v1/terminal-analysis/source-comparisons.csv | Conditional source comparisons |
| production/chapter6-v1/terminal-analysis/condition-summary.csv | All parameter means, uncertainty and recovery diagnostics |
| production/analysis-plan.json | Estimands, denominator policy and multiplicity families |
| simulation/analyze_production.py | Saved-report analysis |
| simulation/build_production_report.py | Render saved analysis tables |
| simulation/openmx_production.R | Gaussian missing-data model implementation |
| handoff/terminal-reproduction.md | Pinned complete saved-analysis recipe |
| outputs/generator-summary-analysis.json | All generation conditions and reliability/skewness summaries |
| outputs/generator-alpha-sensitivity-analysis.json | Paired raw versus standardized alpha |
| handoff/jags-centered-scientific-evidence.json | Four-chain centered diagnostic and scientific limits |

| Artifact | Scientific role |
|---|---|
| handoff/chapter5-analytical-evidence.json | Identification and likelihood qualifications |
| handoff/chapter6-analytical-evidence.json | Covariance construction and observation links |
| handoff/narrow-analytical-evidence.json | Printed-formula and implemented-kernel checks |
| handoff/scale-invariance-evidence.json | Conditional population statement and finite-data example |
| provenance/runtime-packages.json | Pinned statistical runtime |
| provenance/posterior-runtime-packages.json | Diagnostic package versions |
| paper/references.bib | Editable bibliographic source |

The original thesis PDF and source programs are local-only inputs. Re-extracting their values requires separately obtaining those originals; rebuilding saved tables does not. Repeating the statistical fits additionally requires the pinned R/OpenMx or JAGS runtime, full seed and protocol records, and a fresh output namespace preserving all failures and denominators. No immutable revision is asserted here for the entire later evidence collection.

This manuscript can be rendered from the paper directory with `pdflatex current_standalone.tex`, `bibtex current_standalone`, and two further `pdflatex` passes. The bibliography is editable; all scholarly references below are used in the text.

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